Lecture 6
Smart metering & communications

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“This research was funded by the CETPartnership, under the Joint Call 2023. The CETPartnership’s research projects are co-funded by the European Commission (Grant Agreement No. 101069750) and the national funding organizations listed on the CETPartnership website, Project No. CETP-2023-00493.”

Outline

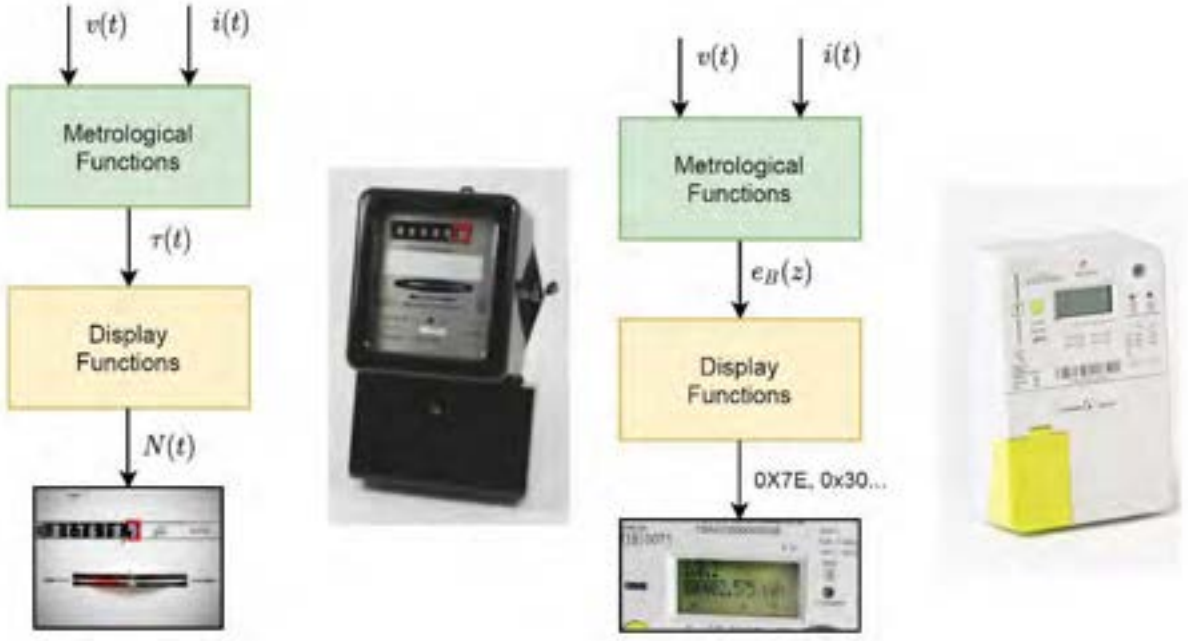
- smart meter functionalities
- smart meter in Flanders
- Atrias as data hub
- broader perspective

Smart meters: potential cornerstone of smart grid

- ferraris meter
  - → replace by digital meter + communication module

- only remote meter reading?
  - advanced monitoring ?
  - control as home energy management system ?
  - link for flexibility provision ?

Analog versus digital meter



(a) Analogue meter (b) Digital meter

Comparison between basic functional blocks of analogue and digital meters

Table 1: Expected benefits and downsides of smart meters for different stakeholders

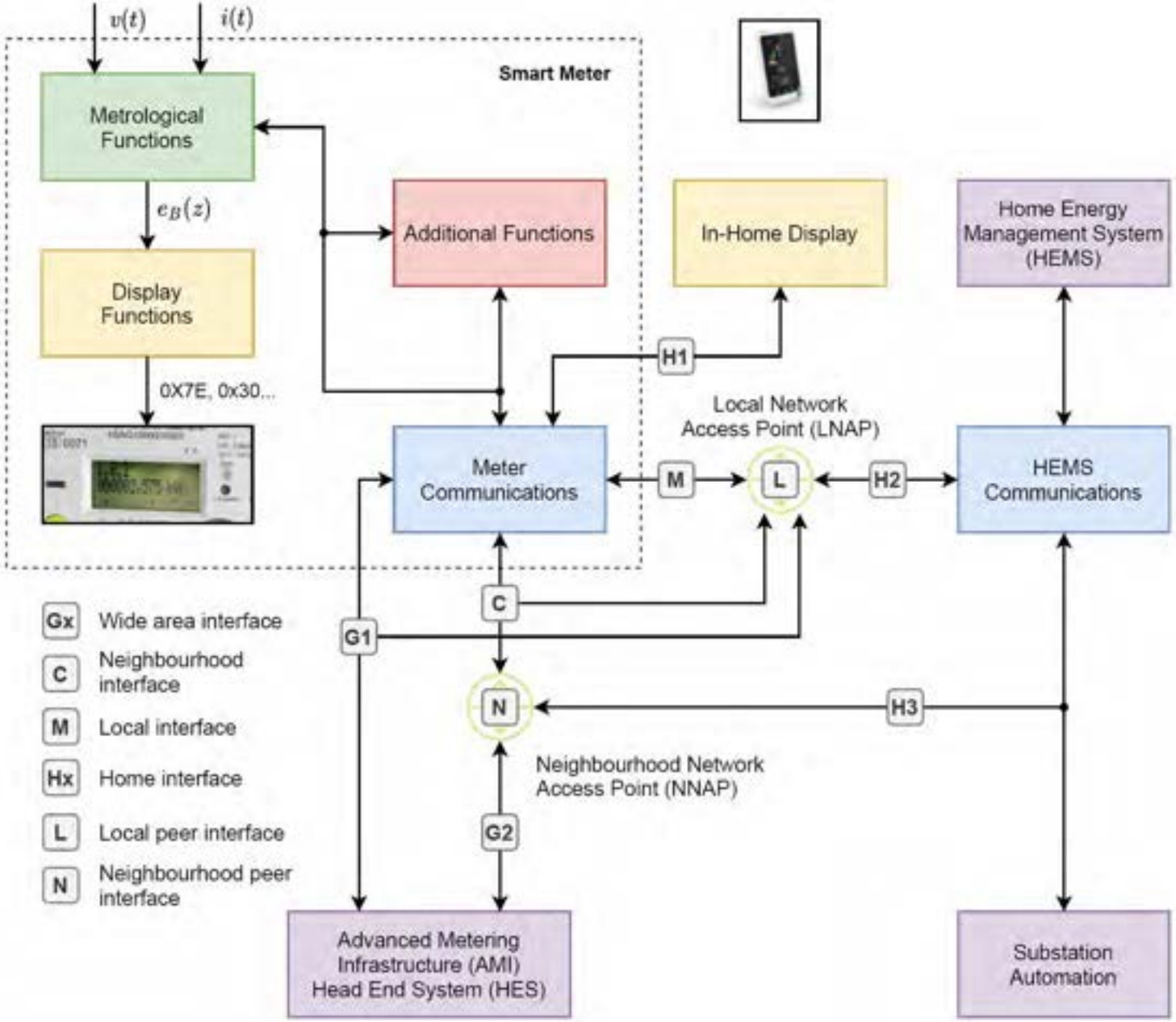
Actor	(Expected) benefits	(Expected) downsides
DSO/TSO	Higher demand flexibility; Reduction of reading costs; Network monitoring; Outages and theft detection	Roll-out cost; Communication infrastructure; Data management infrastructure
Suppliers	Improved information on usage; Improved forecasting; Lower imbalance prices	Higher exposure to cyber-attacks; New training, equipment and production investments
Consumers	Increased energy awareness; Accurate billing; Innovative tariff schemes; Supplier flexibility	Privacy intrusion and data protection concerns; Installation fees

[Premises D1.1, Nov 2022]

Smart metering at EU policy level

- Directive **2006/32/EC** on **energy efficiency and energy services**
  - "…individual meters that accurately reflect the final customer's actual energy consumption and that provide information on actual time of use."
- Directives **2009/72/EC** & **2009/73/EC** on **intelligent metering systems**
  - Member States have to "ensure the implementation of intelligent metering systems that shall assist the active participation of consumers in the electricity/gas supply market…"
- 3/2009: **EC Mandate M/441** to CEN, ETSI, CENELEC on standards for intelligent metering systems
- implementation is up to member states…

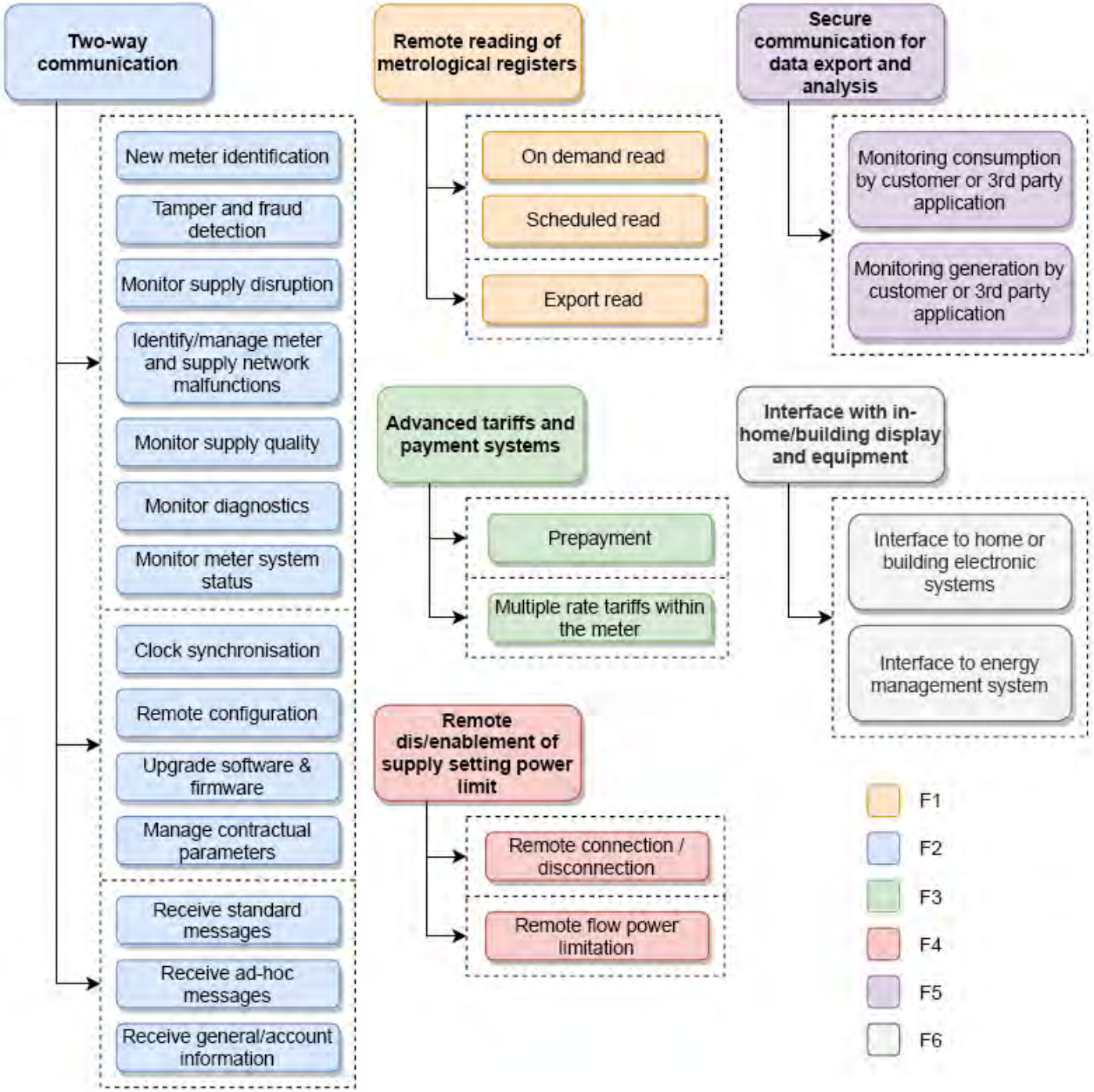
Reference architecture for smart meter communications



[Premises D1.1, Nov 2022]

Smart meter use cases – 6 functional requirements

[Premises D1.2, Nov 2022]



Smart meter functional requirements

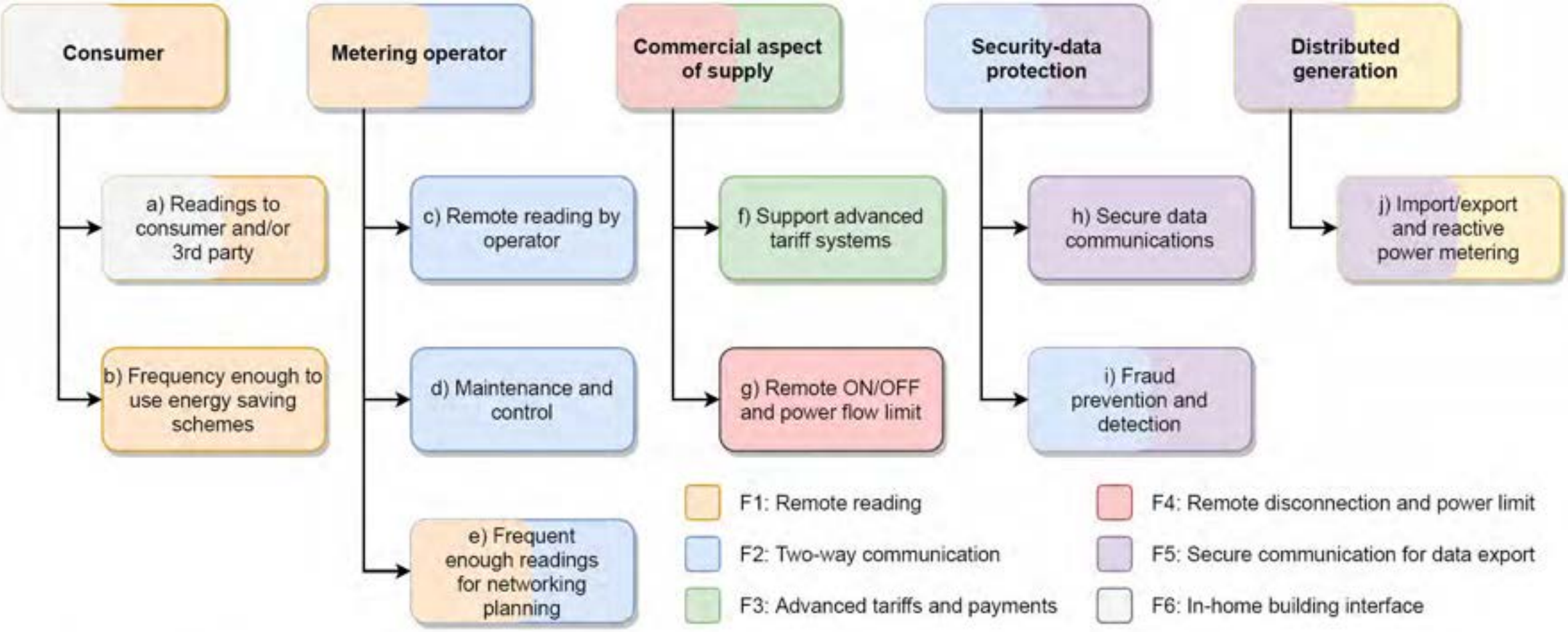
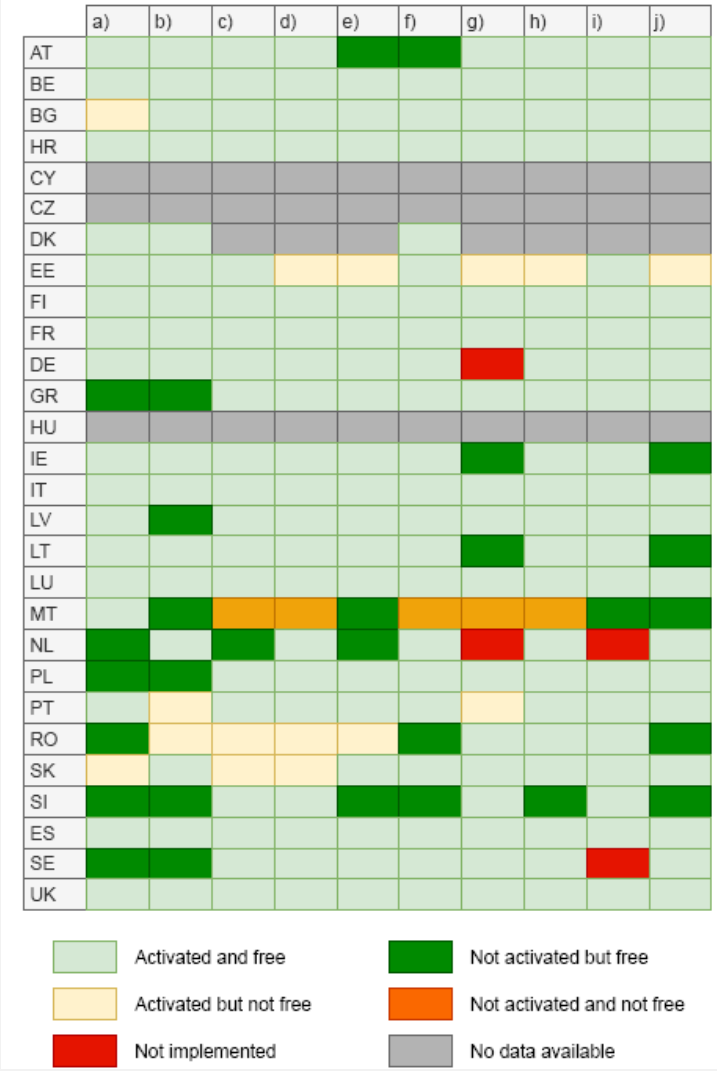


Figure 2: Comparison between CEN/CLC/ETSI/TR 50572:2011 and 2012/148/EU requirements.

[Premises D1.2, Nov 2022]

Functional requirements in different EU countries



[Premises D1.2, Nov 2022]

Which information & communication infrastructure

- wired?
  - PLC (power line carrier) *(Flanders pilot)*
  - others: cable (CATV), phone line (xDSL), ethernet, fibre …
- wireless?
  - cellular: 4G, 5G, LoRa, … *(Flanders deployment: LoRa / narrowband 4G)*
  - others (require intermediate data concentrator): Wi-Fi, ZigBee, RF, …
- choice ?
  - → determines bandwidth, latency, reliability, etc.
  - → determines architecture (direct connection, intermediate infrastructure, shared or dedicated, third party dependence, …)

Smart meter standards according to OSI stack

	Wired Standards			Wireless Standards		
	PLC		Others	Cellular	Others	
Application		AMIS + CX1 OSGP + ETSI-TS 103 908 Meters&More + CLC/50568-4				Kamstrup RF Mesh MeshNet3
Presentation						
Session						
Transport						
Network						
Data Link	IEC 61334-5 (P-LAN) G.9904 (PRIME) G.9903 (G3-PLC) G.9902 (G.hnem) IEEE P1901.2 IEEE P1901 G.9960,61,63 (G.hn) HomePlug AV 2.0		EN 13757-2 (M-Bus) DSMR (IEC 62056-21) G.991-G.993 (xDSL) IEEE 802.3 (Ethernet) G.983-G.984 (Fiber)	GSM-2G GPRS-2.5G EDGE-2.75G UMTS-3G LTE-4G IEEE 802.16 (WiMAX)	EN 13757-4 (wM-Bus) IEEE 802.15.4 IEEE 802.11[x] (Wi-Fi) FlexNet®	
Physical						

Figure 6: Contextualisation of various communication technologies in the OSI model.

[Premises D1.1, Nov 2022]

Technical & non-technical aspects

- all communication means fulfil *minimal* requirements
- choice depends on technical aspects
  - much data → broadband
    - broadband internet-based solutions or cellular 4G
  - real-time → broadcasting
    - PLC or wireless solutions
- *and* non-technical aspects
  - costs
    - PLC/RF and existing internet solutions
  - (in)dependence
    - telecom infrastructure by third parties
- future proof: broadband + real-time + dependable

Outline

- smart meter functionalities
- smart meter in Flanders
- Atrias as data hub
- broader perspective

Smart meters in Flanders

- deployed since July 2019
- target roll-out
  - 2024: 80%
  - 2029: 100%
- access to historical data
  - injection/take-off
    - daily, or (on request) 15-minute data
  - peak power per month

2017 considered scenarios

- scenarios for optimal functionality assignment in *smart meter system*
  - electronic meter without communication
  - **standardised digital meter in advanced, extendible smart meter system**
    - idem, but without direct communication link between meter and regulated back-end
  - advanced (complex) digital meter

[VR 2017 0302 DOC.0062/1QUATER, based on 2016-study EnergyVille for VEA]

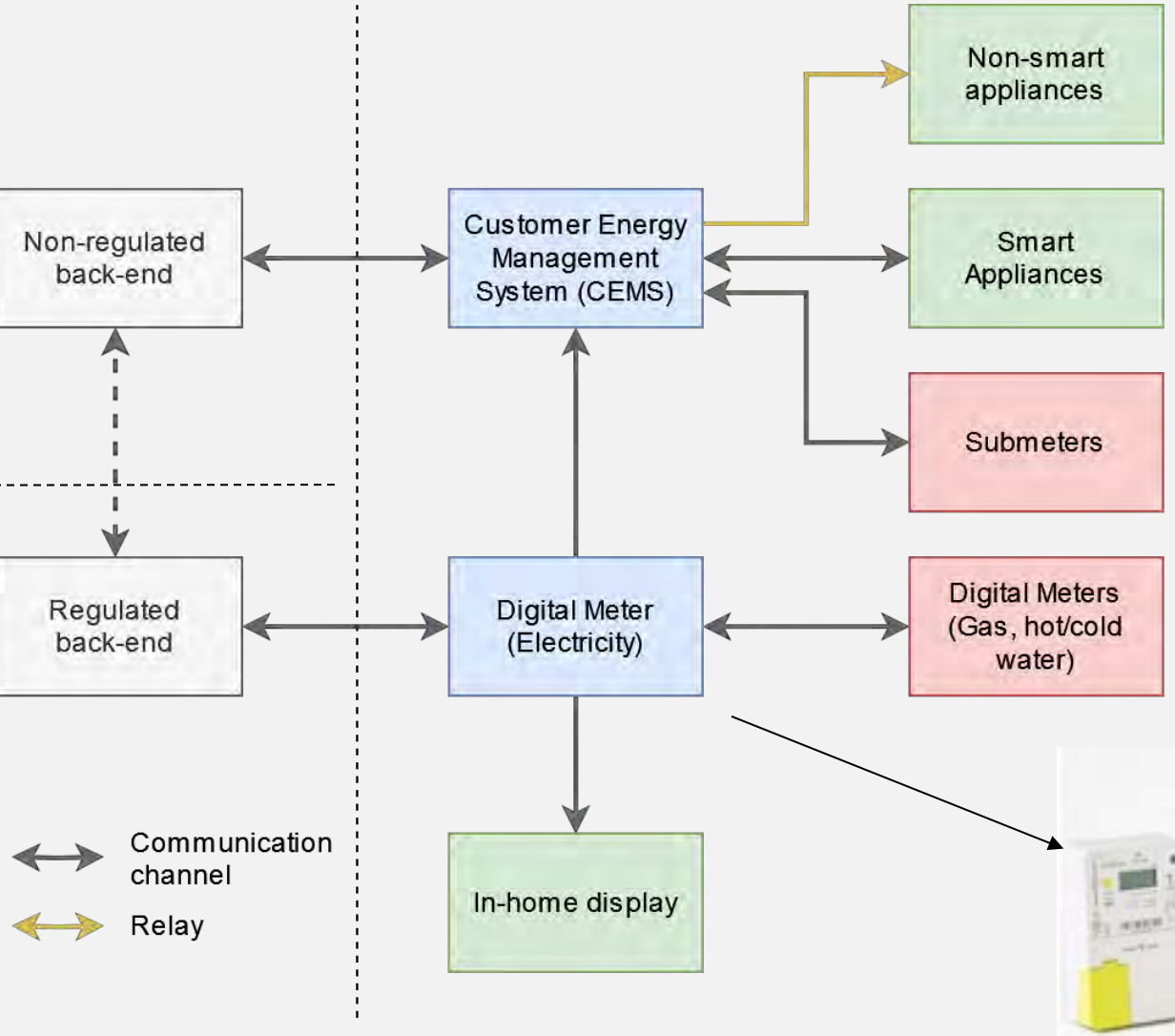
Communication architecture

DSO no control

- via suppliers, ESCOs, dongles
- self-consumption max
- cost minimization
- aggregation (e.g., fleet owners)

- via DSO to central database
- at least 24 h delay

DSO in control

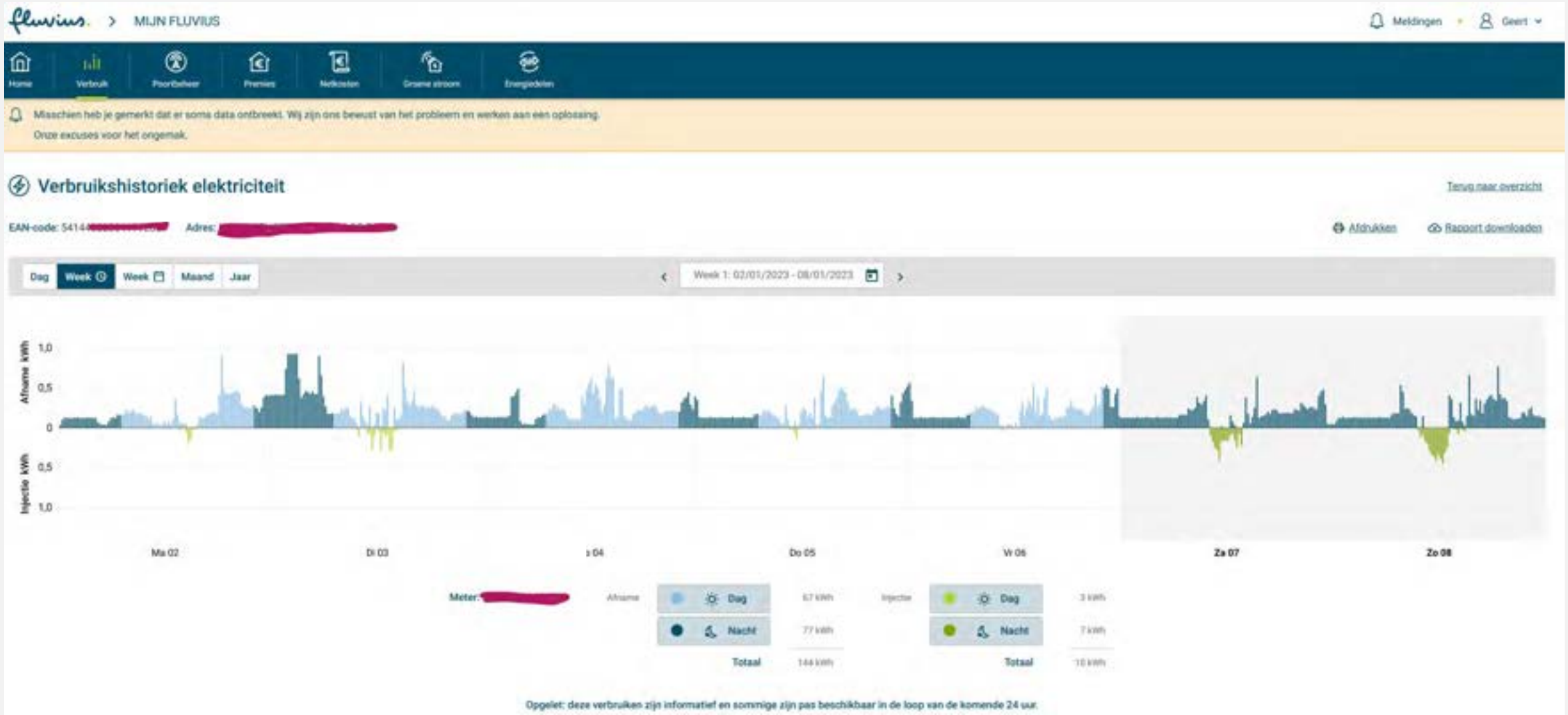


Motivation chosen architecture

[VR 2017 0302 DOC.0062/1QUATER, based on 2016-study EnergyVille for VEA]

- *digital meter* has more added value as part of *smart meter system*
  - with **Customer Energy Management System** (CEMS) and **back-end** functionalities
- cost decrease by moving functionality to back-end or CEMS
  - in this way, budget meters can be implemented without special hardware
- market actors can add functionality, and provide innovative services
  - e.g. for customers from perspective of market (cost), sustainability or comfort
- digital meter must be as standardised as possible
  - all region-specific extensions are expensive
- *digital meter* must be 'lean';
  innovations and additional functionalities *in smart meter system*
  - smart meter system should not be deployed by system operators
  - smart meter system should not be hindered by digital meter
- innovation can be segmented (in time or for different customers)

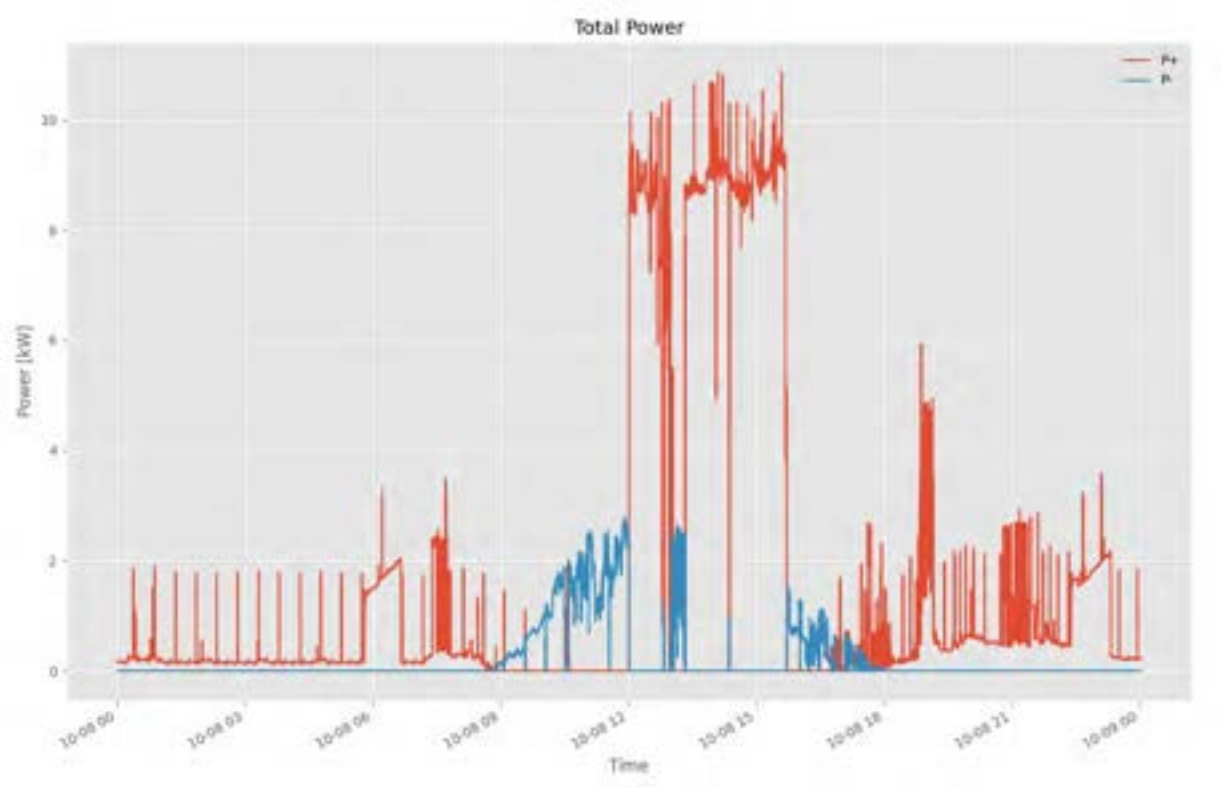
Regulated back-end: kWh@15min take-off/injection



Non-regulated back-end: commercial and research

- Commercial: smappee
  - kW@5min, submetered

- Research: premises
  - kW@1s, per phase



Digital meter > 56 kVA [Fluvius 2023]

What's wrong with Ferraris meter?

- yearly snapshot
  - energy awareness ?
- measure net use of electricity
  - = take-off – injection
- quid photovoltaic panels?
  - net use approximately zero
    - if grid costs based on kWh, then borne by others
  - 2016 fixed contribution to 'energiefonds'
    - proportional to inverter rating

unfair !

unfair !

Capacity tariff, since 1 Jan 2023

more fair !

- component ‘distribution – grid use’
  - based on (peak)power per month
    - 15’ period of each month, with highest average power consumption
    - averaged over last 12 months
    - minimum 2,5 kW
  - measured by digital meter
  - when analogue meter: fixed amount, equivalent to 2.5 kW peak
    - and grid costs based on #kWh
  - when “social tariff”: no capacity tariff
- component ‘distribution’
  - independent of time-of-use period (peak/off-peak)

https://mijn.fluvius.be – peak power

https://mijn.fluvius.be – peak power (home with heat pump) (Jan 2023)

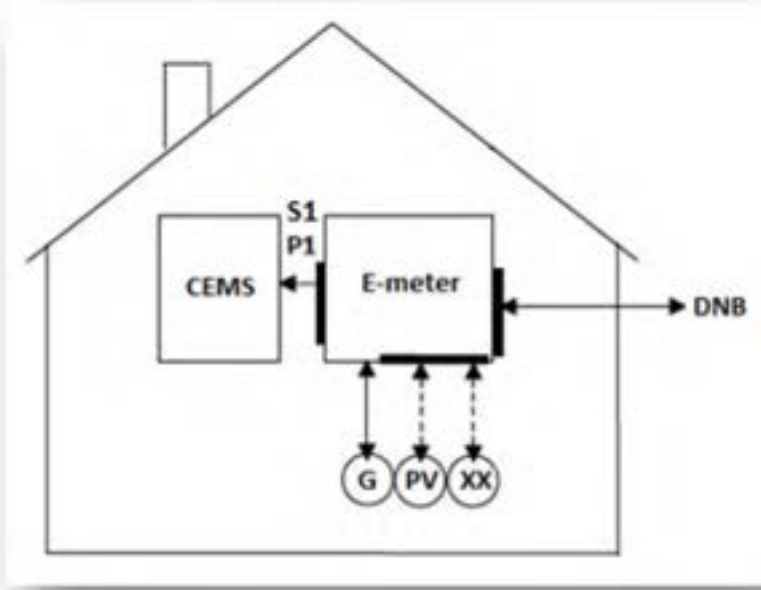
Even more fair, if…

- take local, momentaneous congestion in grid into account…
- → locational, time-dependent pricing…

Smart meter technicalities: port specifications

- specification
  - https://www.fluvius.be/sites/fluvius/files/2019-10/technische-specificaties-gebruikerspoorten-digitale-meters.pdf
- P1 port (and S1) port: link to CEMS (consumer energy management system)
  - local ports to access meter data
- P2 port to other devices
  - gas meter, heat network, others
- P3 port to DSO

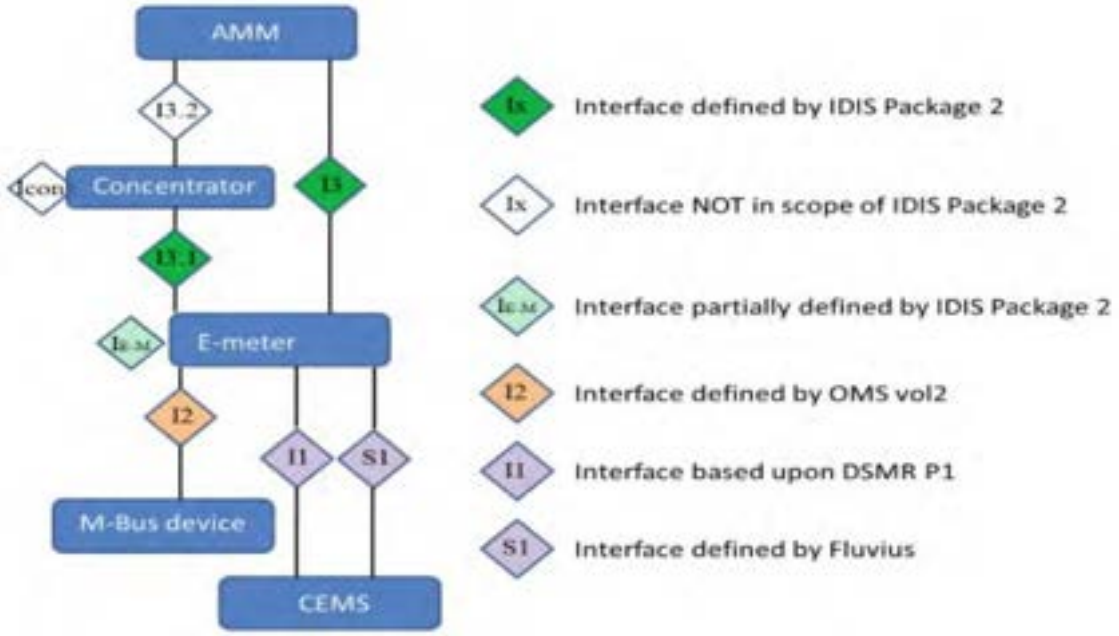
[Fluvius]



Figuur 2: Meetinrichting

Ports of smart meter: standards

- I3/P3 to data system operator
  - based on IDIS Pack 2
  - wireless via Narrowband IoT (4G/LTE)
- I2/P2 to other devices (G-meter, …)
  - based on M-Bus (wired or wireless)
- I1/P1 to CEMS
  - based on DSMR P1, based on OBIS
- S1 to CEMS (2019-2023)
  - defined by Fluvius



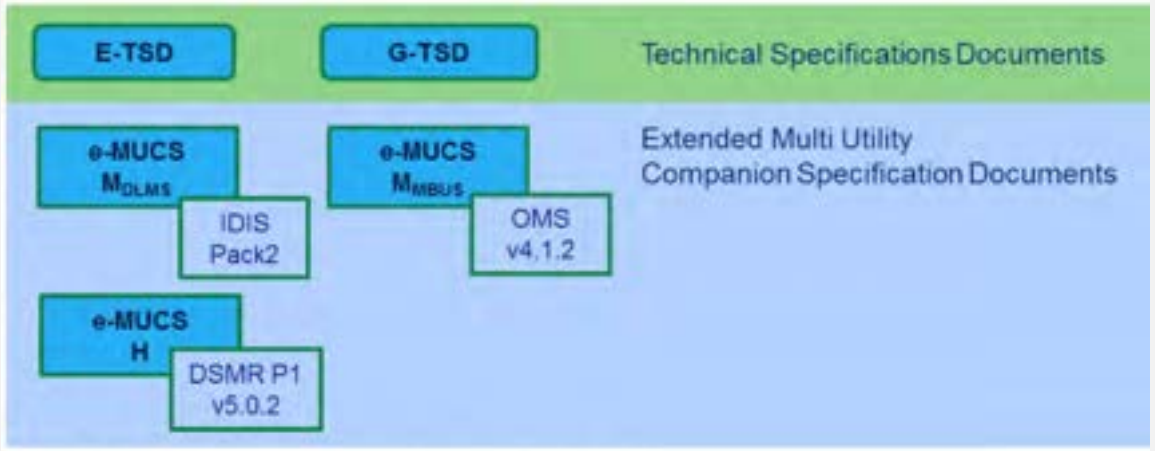


Figure 1 'Set of Requirements' structure of the Smart Metering Solution

- https://www.fluvius.be/sites/fluvius/files/2019-07/1901-fluvius-technical-specification-user-ports-digital-meter.pdf

P1 user port: consumption info (Belgian dataset)

- unidirectional port, every xx minutes
  - serial number E-meter
  - interface version user port
  - timestamp of user port telegram
  - register status of take-off of active energy per counter (E)
  - register status of injected active energy per counter (E)
  - actual active power (take-off)
  - actual active power (injection)
  - active counter (*day/night*)
  - connection status of E-meter
    - connected, disconnected, ready for reconnection
  - threshold for disconnection based on power (3-phase power)
  - threshold for disconnection based on current (per phase)
  - voltage level per phase
  - current level per phase
  - serial number G-meter (if present)
  - register status (index) of G-meter (if present)
  - connection status of G-meter (if present)
  - free text message (not used yet, for info from DSO to user)
  - running peak power average

in 0.001 kWh

1W resolution

in A

in kW

new since 2023

Value	OBIS reference
Equipment identifier	0-0:96.1.1.255
Version information	0-0:96.1.4.255
Date-time stamp	0-0:1.0.0.255
Meter Reading electricity consumption (Tariff 1) in 0,001 kWh	1-0:1.8.1.255
Meter Reading electricity consumption (Tariff 2) in 0,001 kWh	1-0:1.8.2.255
Meter Reading electricity injection (Tariff 1) in 0,001 kWh	1-0:2.8.1.255
Meter Reading electricity injection (Tariff 2) in 0,001 kWh	1-0:2.8.2.255
Actual electricity power consumption (+P) in 1 Watt resolution	1-0:1.7.0.255
Actual electricity power injection (-P) in 1 Watt resolution	1-0:2.7.0.255
Tariff indicator (electricity)	0-0:96.14.0.255
Breaker state (elektricity)	0-0:96.3.10.255
Limiter threshold in kW	0-0:17.0.0.255
Fuse supervision threshold L1 in A	1-0:31.4.0.255
Device-Type (Gas)	0-n:24.1.0.255
M-Bus Device ID 1 (Gas)	0-n:96.1.0.255
Last value of not temperature corrected gas volume in m³, including decimal values and capture time (Gas)	0-n:24.2.3.255
Breaker state (gas)	0-n:24.4.0.255
Text message (max 1024 characters)	0-0:96.13.0.255

Table 6: Implementation of the Belgian minimum dataset for the I1 interface

P1 port: consumption info (Flemish dataset, Fluvius)

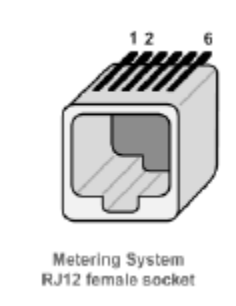
A.2 Flemish dataset (Fluvius)

The Flemish government and regulator have defined extra objects on top of the Belgian minimal dataset. These extra objects can be found in this section. This addition on top of the Belgian minimal dataset will be implemented on the meters for the Flemish DSO's

Value	OBIS reference
Instantaneous voltage L1	1-0:32.7.0.255
Instantaneous voltage L2	1-0:52.7.0.255
Instantaneous voltage L3	1-0:72.7.0.255
Instantaneous current L1	1-0:31.7.0.255
Instantaneous current L2	1-0:51.7.0.255
Instantaneous current L3	1-0:71.7.0.255

P1 port: physical interface

- as in NL: DSMR 5.0.2 P1
  - https://www.netbeheernederland.nl/_upload/Files/Slimme_meter_15_a727fce1f1.pdf
  - RJ12 connector
    - incl. 5V DC



Pin #	Signal name	Description	Remark
1	+5V	+5V power supply	Power supply line
2	Data Request	Data Request	Input
3	Data GND	Data ground	
4	n.c.	Not connected	
5	Data	Data line	Output. Open collector
6	Power GND	Power ground	Power supply line

Table 5-1: Physical connector pin assignment

  - galvanic isolation

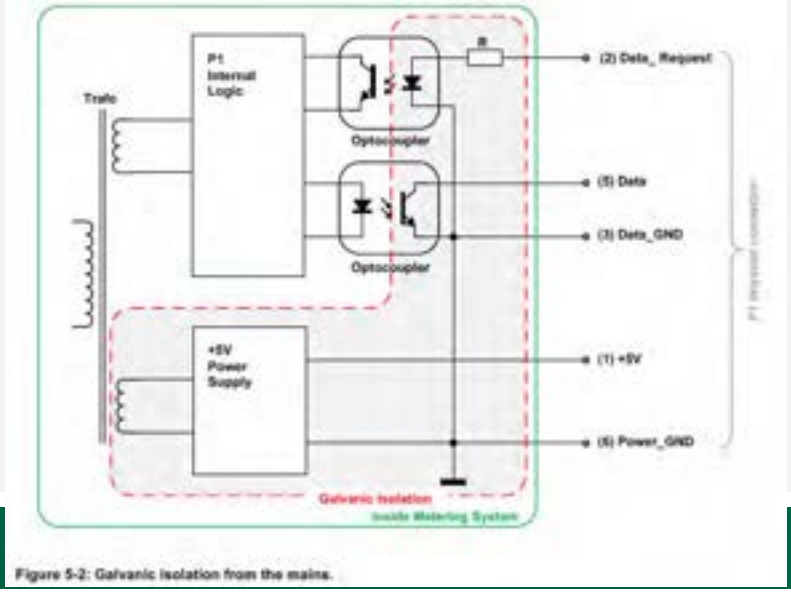
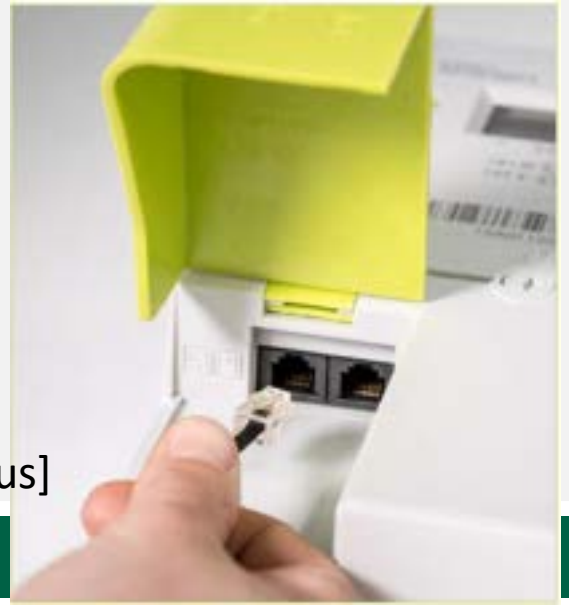


Figure 5-2: Galvanic isolation from the mains.



[Fluvius]

P1 port: software interface

- partially based on (NL-) DSMR 5.0.2 P1 , but with different data fields
  - → e-MUCS – H

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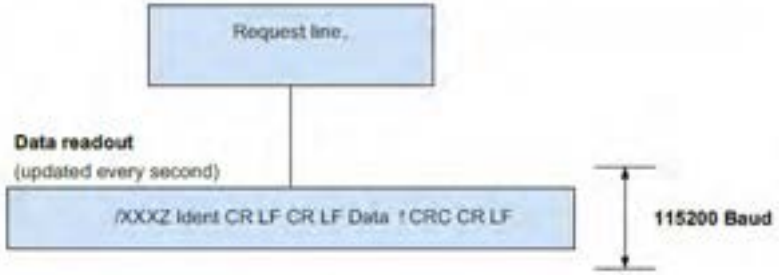
6 PROTOCOL DESCRIPTION

The protocol is based on NEN-EN-IEC 62056-21 Mode D. Data transfer is requested with request line and automatically initiated every second until request line is released. The information in the P1 telegram must be updated every second.



6.1 Transfer speed and character formatting

The interface must use a fixed transfer speed of 115200 baud.

The Metering System must send its data to the OSM device every single second and the transmission of the entire P1 telegram must be completed within 1s.

The format of transmitted data must be defined as "8N1". Namely:
- 1 start bit,
- 8 data bits,
- no parity bit and
- 1 stop bit.

Note:
This is not conforming to EN-IEC 62056-21 Mode D – Ref [1].

6.2 Data readout

The Metering System transmits the data message, as described below, immediately following the activation through the Request signal.

/	X	X	X	5	Identification	CR	LF	CR	LF	Data	!	CRC	CR	LF

One of the COSEM object attribute values in the data blocks could be used for firmware updates or parameter update of a P1 device. Because of this it is not allowed to split the large data block (which can contain up to 1024 characters).

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6.13 Example P1 telegram

The following table shows an example of the P1 telegram that is in accordance to IEC 62056-21 Mode D.

Be aware of the fact that the number of OBIS codes and the order of OBIS codes is not fixed. The P1 device must be able to interpret the OBIS codes and understand the representation. For example: depending on installation, the OBIS codes for a heat/cold meter might be sent before the OBIS codes of a Gas meter.

The example telegram below is based on:

- DSMR version 5.0 (value 50 behind OBIS code 1-3:0.2.8.255)
- It is send at 2010, December 9th, 11h30m20s
- Gas value of 2010, December 9th, 11:25h is presented
- 4 power failures in any phase
- 2 long power failure in any phase
- Power Failure Event log:
  Failure at 2010, December 8th, 15h20m15s, duration 240 seconds
  Failure at 2010, December 8th, 15h05m03s, duration 301 seconds
- 2 voltage sags in phase L1
- 1 voltage sag in phase L2 (poly phase meters only)
- 0 voltage sags in phase L3 (poly phase meters only)
- 0 voltage swells in phase L1
- 3 voltage swells in phase L2 (poly phase meters only)
- 0 voltage swells in phase L3 (poly phase meters only)
- Only one M-Bus device is connected to the Electricity meter. The register value of the Gas meter is 12785,123 m^3. This value is captured by the G meter at 2010, December 9th,at 11h25m00s Wintertime.
- Instantaneous voltage per phase
- Instantaneous current per phase
- Instantaneous active power (+P) per phase
- Instantaneous active power (-P) per phase

```
/ISk5\2MT382-1000

1-3:0.2.8(50)
0-0:1.0.0(101209113020W)
0-0:96.1.1(4B384547303034303436333935353037)
1-0:1.8.1(123456.789*kWh)
1-0:1.8.2(123456.789*kWh)
1-0:2.8.1(123456.789*kWh)
1-0:2.8.2(123456.789*kWh)
0-0:96.14.0(0002)
1-0:1.7.0(01.193*kW)
1-0:2.7.0(00.000*kW)
0-0:96.7.21(00004)
0-0:96.7.9(00002)
1-0:99.97.0(2)(0-0:96.7.19)(101208152415W)(0000000240*s)(101208151004W)(0000000301*s)
1-0:32.32.0(00002)
1-0:52.32.0(00001)
1-0:72.32.0(00000)
1-0:32.36.0(00000)
1-0:52.36.0(00003)
1-0:72.36.0(00000)
0-
0:96.13.0(303132333435363738393A3B3C3D3E3F303132333435363738393A3B3C3D3E3F303132333435363738393A3B3C
3D3E3F303132333435363738393A3B3C3D3E3F303132333435363738393A3B3C3D3E3F)
1-0:32.7.0(220.1*V)
1-0:52.7.0(220.2*V)
1-0:72.7.0(220.3*V)
1-0:31.7.0(001*A)
1-0:51.7.0(002*A)
1-0:71.7.0(003*A)
1-0:21.7.0(01.111*kW)
1-0:41.7.0(02.222*kW)
1-0:61.7.0(03.333*kW)
1-0:22.7.0(04.444*kW)
1-0:42.7.0(05.555*kW)
1-0:62.7.0(06.666*kW)
0-1:24.1.0(003)
0-1:96.1.0(323232324142434431323334353637383 9)
0-1:24.2.1(101209112500W)(12785.123*m3)
!EF2F
```

IEC 62056-21 - DSMR 5.0 P1 companion

- protocol: IEC 62056-21 Mode D, with exceptions
  - 1-way, simplex
  - data transfer requested with request line
    - automatically initiated every second until request line released
  - fixed transfer speed of 115200 baud, instead of 2400 baud
- data:
  - uses reduced OBIS (OBject Identification System) identification code
  - message according IEC 62056-21
    - but own COSEM/OBIS data representation format

IEC 62056-21 - DSMR 5.0 P1 companion

- electrical interface IEC 62056-21, RJ12 connector
  - simple installation by customer
  - low cost
  - safe for the customer
  - metering system or the data in it cannot be compromised
- extra +5V power supply, up to 250 mA
  - is meant to provide a power source to HAN devices, to enable their ability to process and transfer received metering data further wired or wireless
    - i.e. via Bluetooth or Wi-Fi technologies

S1-interface (Fluvius) – *(only 1st generation 2019-2023)*

- S1: designed to provide "raw" data CEMS at a particularly high frequency
  - CEMS application must further process this "raw" data to make advanced calculations allowing for very detailed consumption feedback and control.
  - can be used to enable device detection (*Non-Intrusive Load Monitoring*).
  - "raw" data provides developer with access to very detailed information on voltage and current consumption on the connection. Combining the consumption feedback from the P1 user port with additional data from the S1 user port allows for advanced control options.
- only data relating to electricity meter (no submeters)
  - Serial number of the electricity meter
  - Sampling of the sine-wave voltage between phase 1 and neutral
  - Sampling of the sine-wave voltage between phase 2 and neutral (only on polyphase meters)
  - Sampling of the sine-wave voltage between phase 3 and neutral (only on polyphase meters)
  - Sampling of the sine-wave current in phase 1
  - Sampling of the sine-wave current between phase 2 and neutral (only on polyphase meters)
  - Sampling of the sine-wave current between phase 3 and neutral (only on polyphase meters)
  - Sampling of the sine-wave current in neutral
- sampling frequency 2600 Hz, but can be changed in future (2000-4000 Hz initially)

Prepayment meter Fluvius (“budgetmeter”)

- old (analogue)
  - meter replacement necessary
  - uses prepayment card
  - precharging only at specific locations
  - no view on daily usage
  - no info about balance *(saldo)*
    - except via screen on meter

- new (digital) – *computation in back-end*
  - ordinary digital meter
    - no replacement necessary
    - same look = less stigmas
  - no cards necessary
  - payments 24/7
    - no displacement needed
  - balance & usage via internet or freephone
  - info via text message / email

[Fluvius]

Digital meter privacy framework

- customers own the data
  - explicit consent required for 3rd parties
- DSO accesses data to fulfil legal tasks
  - one measurement/day
    - detailed values ‘later’
  - Power Quality analysis
    - limited time, aggregated, anonymised
- users’ ports (P1/S1) deactivated by default
- easy access/parsing with DSMR standard and self-developed open-source tool

fluvius.

Outline

- smart meter functionalities
- smart meter in Flanders
- Atrias as data hub
- broader perspective

www.atrias.be

Home | About Atrias | Jobs | Current Documentation | Projects | Onboarding | FAQ | EN

About Atrias

Atrias provides support for the development of the liberalised Belgian energy market on the distribution level, playing a key role as an information provider. Atrias cv is a joint initiative of the four largest distribution system operators in Belgium, i.e., Fluvius, ORES, Sibelga and RESA.

The company was established on 9 May 2011, with registered office at 17A, Kanselarijstraat, 1000 Brussels.

Our vision

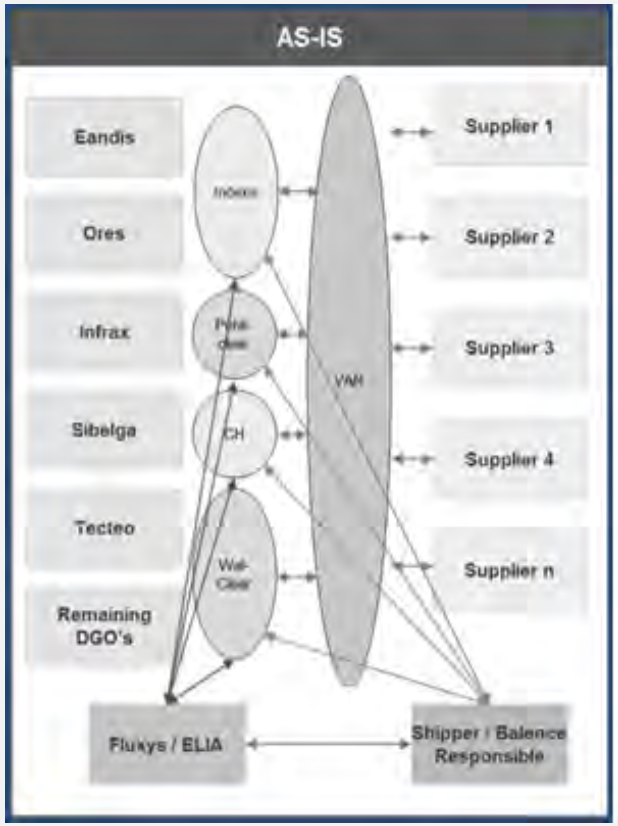
Atrias is the reference in terms of knowledge and expertise of market processes and standards. We provide reliable and high-quality services on centralised, federal platforms.

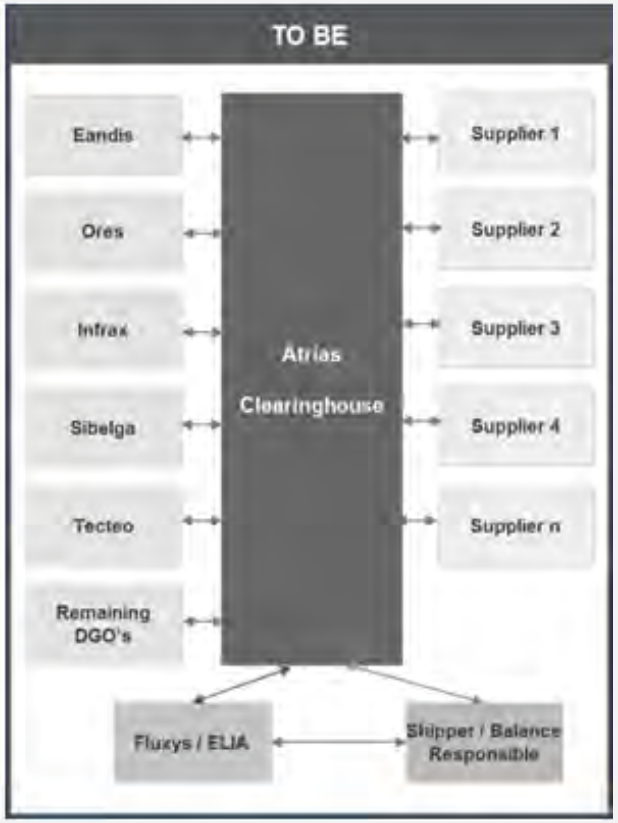
What does Atrias do?

We manage the market facilitation documentation for and in close consultation with the distribution system operators. We also develop a reliable and innovative IT infrastructure, that benefits all the market players. The systems that are centrally managed by Atrias result in substantial cost savings for society, offering the necessary flexibility for future evolutions.

MIG 6 – new market model and data exchange

- Jan. 2018 MIG 6.0 (market implementation guide)
  - federal clearing house
- Jan. 2021: MIG 6.5.1.31
  - ‘for implementation’
- started end 2021





MIG 6

- MIG 6 introduces
  - new data model
    - with Service Delivery Points instead of EAN-numbers
  - new message format
    - XML based
  - new protocols for communication among market participants
    - based on web services
- 40 existing (MIG 4.1) scenarios replaced by 15 modules and labels
  - new functionalities, concepts and processes
    - to support smart meters
    - e.g. *preswitching* between suppliers

[atrias.be]

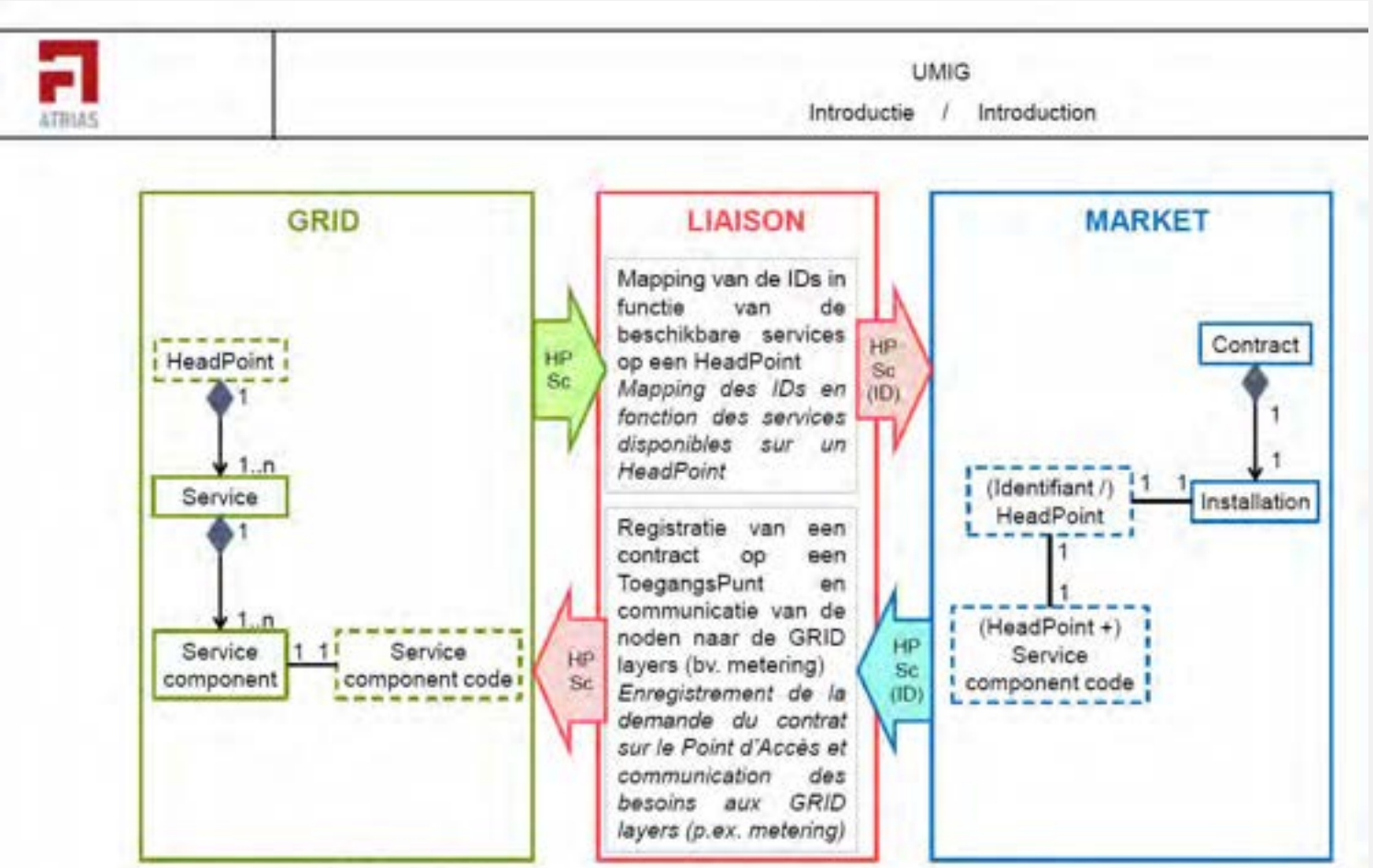


Figure 3 - Model v1.0

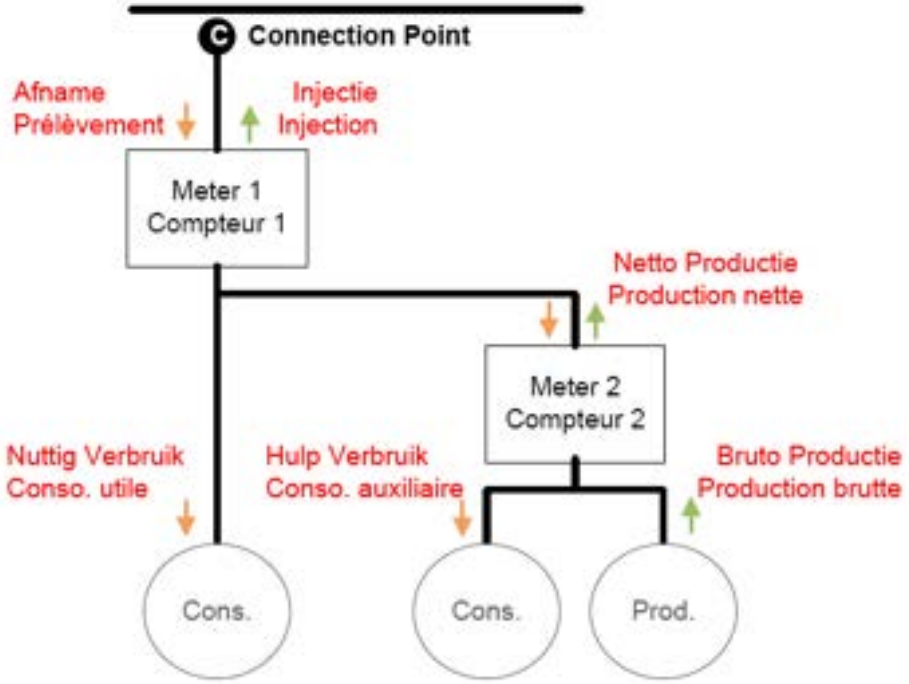
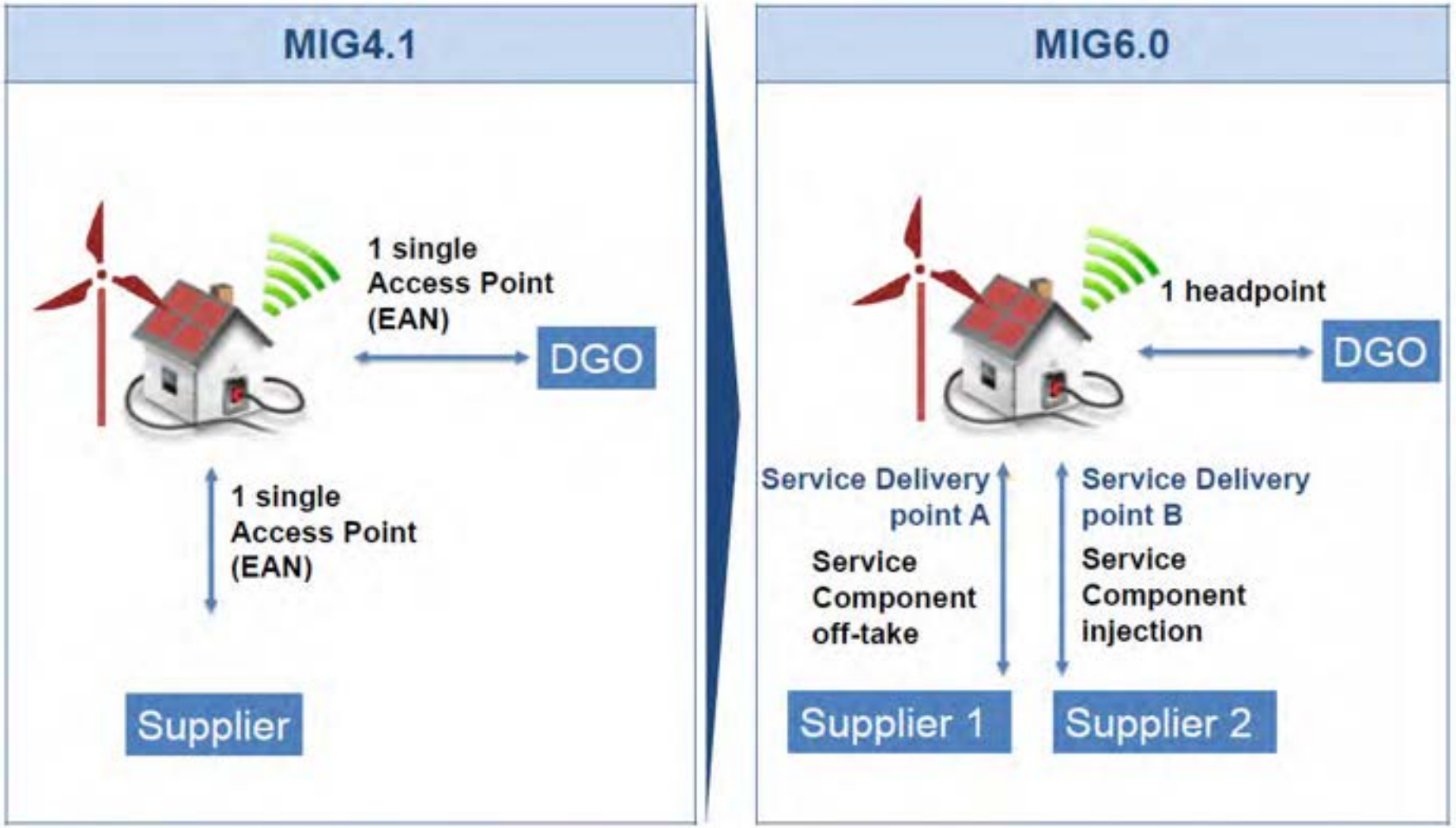


Figure 4 - Local production v1.0

MIG6.0: Key changes voor leveranciers

New Data Model

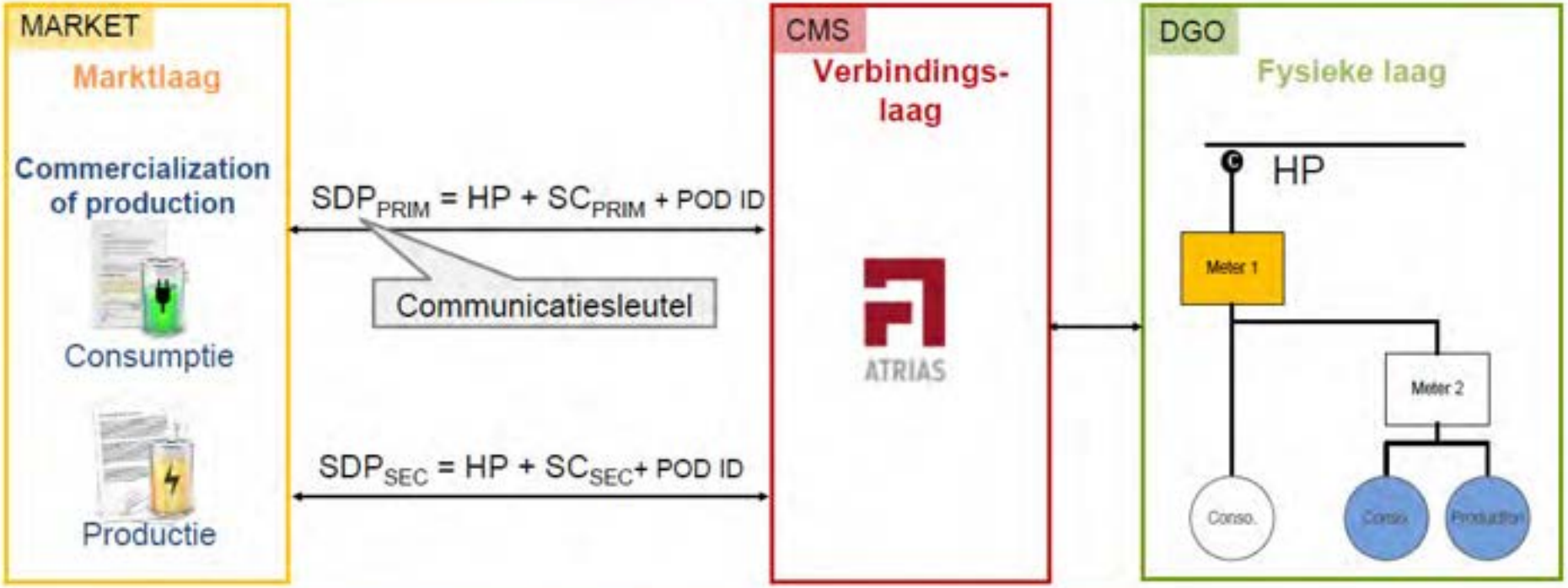


MIG6.0: Key changes voor leveranciers

New Data Model

ATRIAS

SDP	=	HP	+	SC	(+POD ID)
↓		↓		↓	↓
ServiceDeliveryPoint	=	Headpoint	+	ServiceComponent	(+PointOfDeliveryID)
↓		↓		↓	↓
Kleinste switchable eenheid (=contract)		Klantcentraal (uniek)		Klant service	Technische supplier key (optional)



https://www.atrias.be/current-documentation

Current Market model

Do you wish to know which processes are used to exchange information between the various market roles on the distribution level in the energy market? Do you want to learn more about the messages that are used for this as a developer of IT systems?

You can find all the relevant information in this section. This includes information about the processes and messages th are currently used. Future versions can be found in the project documentation section.

MARKET PROCESSES | SUPPORTING DOCUMENTS | ARCHIVES

General

General

Name	Modified
UMIG - GE - KD - 01 - Glossary v6.5.1.5.xlsx	20/06/2022
UMIG - GE - KD - 01 - Introduction v6.5.1.5.pdf	20/06/2022
UMIG - GE - KD - 01 - Market Roles v6.5.1.5.pdf	20/06/2022
UMIG - GE - KD - 04 - Input Volumes Processes v6.5.1.31.xlsx	20/06/2022

Business Requirements

Business Requirements

Name	Modified
UMIG - BR - BI - 02 - Grotte Billing v6.5.1.31.pdf	20/06/2022
UMIG - BR - BI - 02 - Public Service Obligation (Brussels) v6.5.1.28.pdf	20/06/2022
UMIG - BR - BI - 02 - Transaction Costs (Brussels) v6.5.1.28.pdf	20/06/2022
UMIG - BR - ME - 02 - Meterreporting - Evaluation Pro v6.5.1.30.xlsx	20/06/2022
UMIG - BR - ME - 02 - Measure Process v6.5.1.30.pdf	20/06/2022
UMIG - BR - ME - 03 - Exchange Metering v6.5.1.31.pdf	20/06/2022
UMIG - BR - ME - 03 - Request Metering v6.5.1.28.pdf	20/06/2022
UMIG - BR - ME - 04 - Interface Description v6.5.1.28.xlsx	20/06/2022
UMIG - BR - OP - 02 - Operational Gas v6.5.1.13.pdf	20/06/2022
UMIG - BR - SE - 02 - Allocation Process Electricity v6.5.1.31.pdf	20/06/2022

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Information Models

Information Models

Name	Modified
UMIG - IM - BI - 05 - Bill Exchanged Information v6.5.2.1.pdf	20/06/2022
UMIG - IM - BI - 05 - Message Details v6.5.2.1.xlsx	20/06/2022
UMIG - IM - ME - 05 - Measure Exchanged Information v6.5.2.1.pdf	20/06/2022
UMIG - IM - ME - 05 - Message Details v6.5.2.2.xlsx	20/06/2022
UMIG - IM - OP - 05 - Exchanged Information v6.5.1.17.pdf	20/06/2022
UMIG - IM - OP - 05 - Message Details v6.5.1.17.xlsx	20/06/2022
UMIG - IM - SE - 05 - Exchanged Information v6.5.1.26.pdf	20/06/2022
UMIG - IM - SE - 05 - Message Details v6.5.1.31.xlsx	20/06/2022
UMIG - IM - ST - 05 - Exchanged Information v6.5.2.2.pdf	20/06/2022
UMIG - IM - ST - 05 - Message Details v6.5.2.2.1.xlsx	14/11/2022

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XML Schemas

XML Schemas

Name	Modified
UMIG - IM - KD - 06 - Versioning of XSDs and WSDLs v6.5.1.30.pdf	20/06/2022
UMIG - IM - KD - 06 - XML Schemas v6.5.2.2.zip	20/06/2022

Publication Overview

Publication Overview

Name	Modified
Inventory Publications UMIG v6.5.2.2.1.xlsx	14/11/2022
Release note - UMIG v6.5.2.2.1.xlsx	30/06/2022

Outline

- smart meter functionalities
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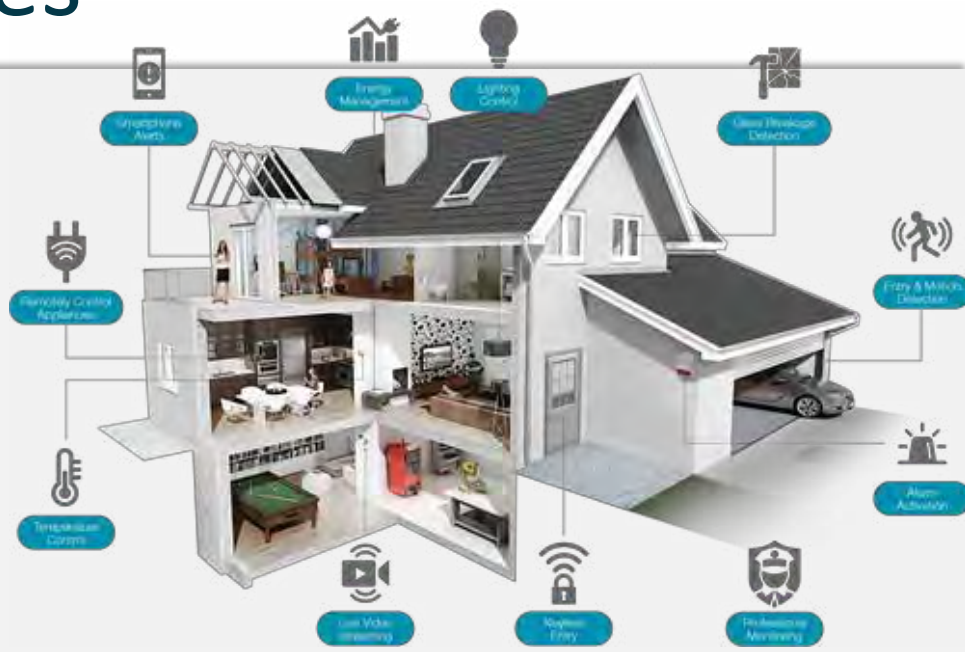
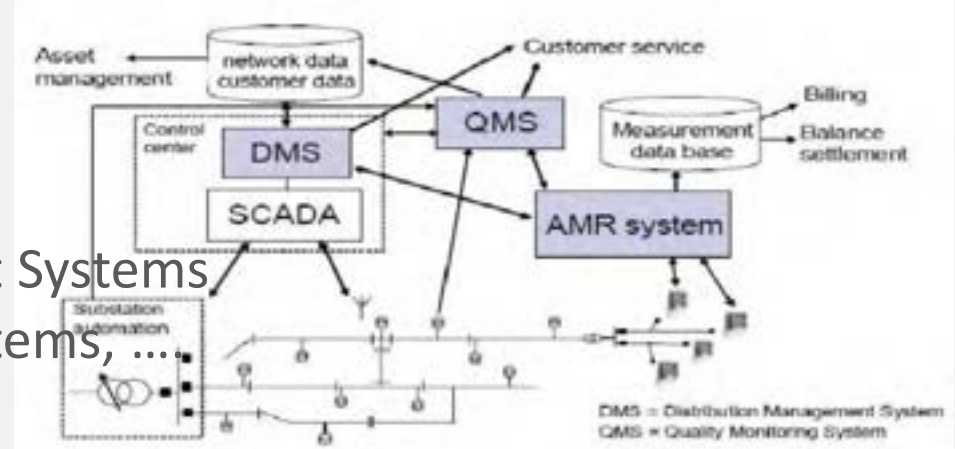
Next generation smart meter services

- services for the customer
  - full integration with Demand Response services
  - full integration with smart home functionality
  - market integration for prosumers
  - tailored retail contracts, ….
- services for energy service companies
  - energy declarations
  - energy consultations
  - aggregation of end customers
  - electric Vehicle charging support/control, ….
- services for distribution system operator
  - integration of customer data in Network Planning Tools
  - integration of customer data in Distribution Management Systems
  - integration of customer data in Outage Management Systems, ….

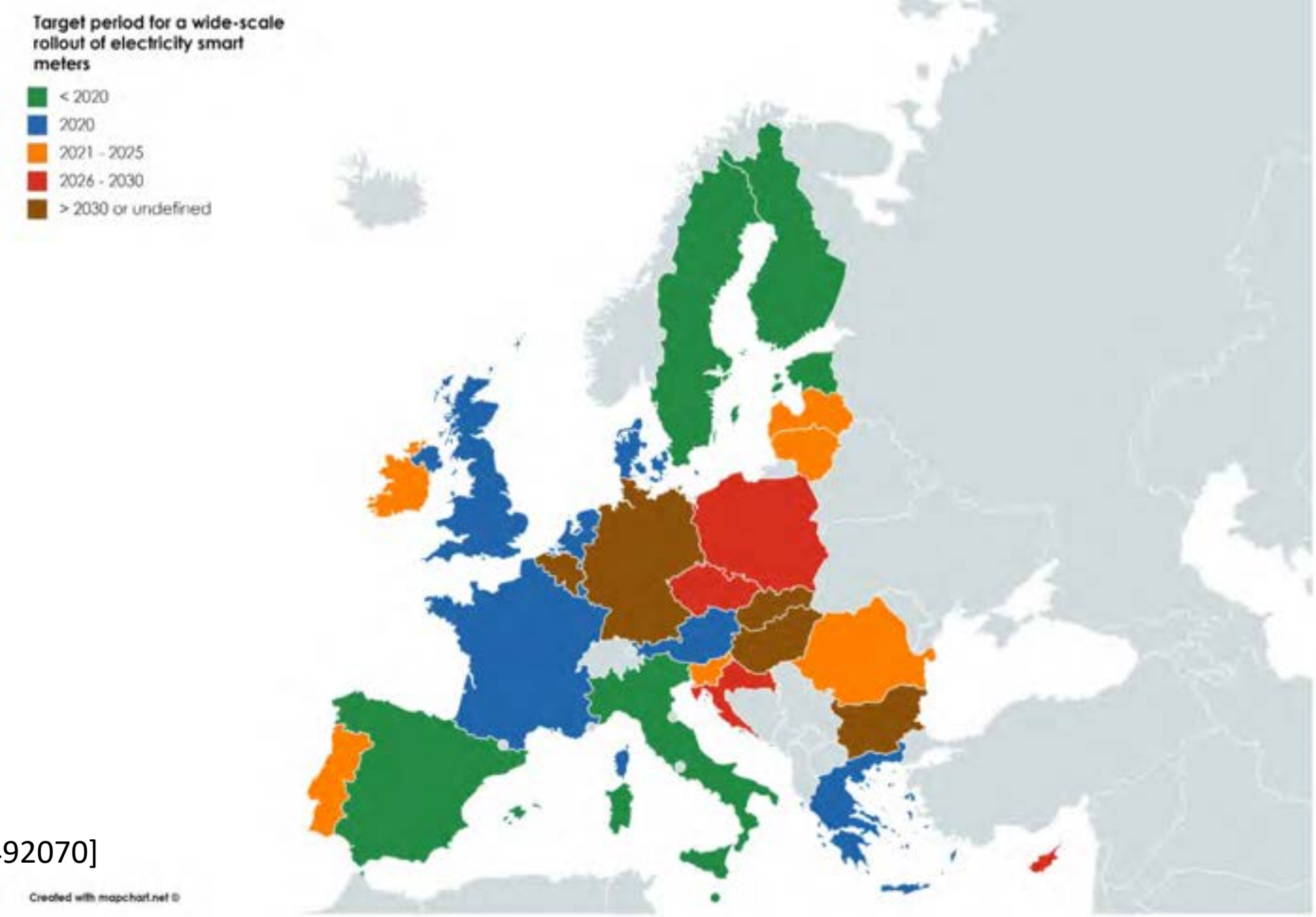




[Johan Söderbom, RISE]

Background: smart metering in EU

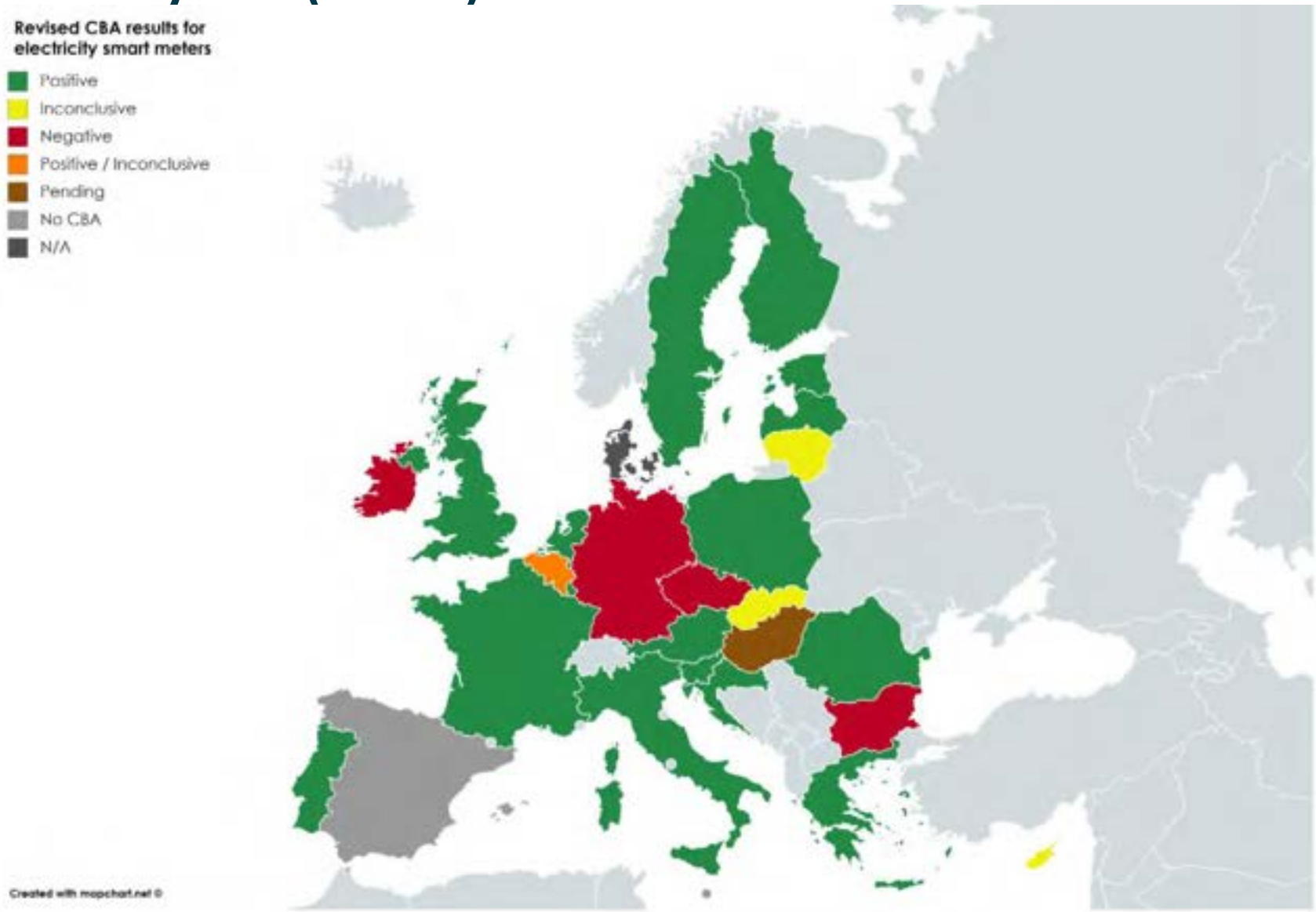
- European Commission, Directorate-General for Energy, Alaton, C., Tounquet, F., Benchmarking smart metering deployment in the EU-28 : final report, Publications Office, 2020, https://data.europa.eu/doi/10.2833/492070



[DOI 10.2833/492070]

Figure 3: Overview of target period for a wide-scale rollout of electricity smart meters with at least 80 % of all consumers for each Member State.

Cost-benefit analysis (CBA)



[DOI 10.2833/492070]

Figure 1: Revised CBA results electricity smart meters considering a large-scale rollout to at least 80% by 2020 (as of July 2018)

CBA – key parameters

Smart metering deployment for electricity

	Range of value	Average based on data from positively assessed cases[4]
Lifetime	8 to 50 years	15 ± 4 years (57%)[5]
Evaluation period	8 to 50 years	23 ± 13 years (69%)
Costs per metering point	€38 to €546	€202 ± €108 (68%) Weighted average[6]: € 172
Benefits per metering point	€44 to €551	€271 ± €143 (62%) Weighted average[6]: €253
Energy savings as reported in pilots[7]	5.42% - 7.85%[8]	N/A

Table 1: Key electricity CBA parameters

[DOI 10.2833/492070]